

Decimal Division ($\div$ Integer + $\div$ Decimal) + Remainder Placement

Unit 1 · Decimal Operations · 65 min · 1-to-3 Online Lesson

Textbook Alignment: New Thinking in Primary Mathematics (2nd Ed.) P6 Book A, Unit 1 + Modern Education P6A Unit 1

Core Traps: T2 Decimal Division — Forgetting to restore the remainder's decimal point after shifting the divisor · Misplacing the decimal point in the quotient

SSPA Relevance: High Frequency Tested every year, accounts for approximately 12–18% of Paper 1

Prerequisites: P5 Decimal Multiplication (Lessons 6–7) · Integer Division (P4) · Decimal Addition & Subtraction (P4–P5)

Lesson Objectives: Decimal $\div$ Integer · Decimal $\div$ Decimal (synchronised shifting) · Remainder decimal point placement · Application problems

Student Name:

Class:

Date:

Duration:

I. Warm-Up Questions (5 questions, 5 minutes)

#	Question	Difficulty	Working (show full working steps)
1	Calculate $48 \div 6 = ?$	Basic	
2	Calculate $125 \div 5 = ?$	Basic	
3	Calculate $3.6 + 2.4 = ?$ (P5 decimal addition review)	Basic	
4	Calculate $5.8 - 2.3 = ?$ (P5 decimal subtraction review)	Basic	
5	Express \$4.50 as "___ dollars ___ cents"	Advanced	

II. Core Knowledge + Worked Examples

KP1: Decimal $\div$ Integer SSPA

Calculation Steps (column method):

The decimal point in the quotient must **align with the decimal point in the dividend**.

If there is a remainder, append "0" to the remainder and continue dividing.

When the dividend digit is smaller than the divisor, write "0" in the quotient — never skip a place.

Key memory point: "Quotient decimal point = dividend decimal point"

Trap-Detonating Example (the most important demonstration of this lesson)

Calculate: $7.5 \div 5 = ?$

Common Error (60% of students)

$$7.5 \div 5 = 15$$

Forgot to place the decimal point in the quotient — $7.5 \div 5 = 1.5$, not 15

Correct Method

$$7.5 \div 5 = 1.5$$

$5 \times 1 = 5$, remainder 2.5; $5 \times 0.5 = 2.5$, quotient = 1.5

Mnemonic: "The quotient's decimal point aligns with the dividend's; if there is a remainder, append a zero and keep dividing."

(Cantonese original: 「商的小數點，對齊被除數；餘數補個零，繼續除落去」)

Most Frequent Error: Forgetting to place the decimal point in the quotient. Remember: $7.5 \div 5 = 1.5$, NOT 15!

Second Most Frequent Error: Forgetting to append 0 to the remainder and continue dividing, leading to an inaccurate quotient.

KP1 Example Practice (show column working)

#	Question	Difficulty	Working
Ex1	$3.6 \div 3 = ?$		
Ex2	$9.2 \div 4 = ?$		

KP2: Decimal ÷ Decimal (Synchronised Shifting Method) SSPA Must-Know

Three-Step Synchronised Shifting Method:
When the divisor is a **decimal**, use the property that multiplying the dividend and divisor by the same number preserves the quotient:
Divisor has 1 decimal place → multiply both numbers by 10
Divisor has 2 decimal places → multiply both numbers by 100
Divisor has 3 decimal places → multiply both numbers by 1000
Essence: Convert "decimal ÷ decimal" into "integer ÷ integer" for calculation.

Example

Ex3: $3.6 \div 1.2 = ?$ (Divisor has 1 decimal place → multiply both by 10: $36 \div 12$)

Example

Ex4: $0.75 \div 0.15 = ?$ (Divisor has 2 decimal places → multiply both by 100: $75 \div 15$)

KP2 Synchronised Practice

#	Question	Difficulty	Working
6	$4.8 \div 0.6 = ?$		
7	$5.4 \div 0.9 = ?$		
8	$2.4 \div 1.2 = ?$		
9	$7.2 \div 2.4 = ?$		
10	$9.1 \div 1.3 = ?$		

KP3: Remainder Decimal Point Placement — "Catastrophe" Special Drill

SSPA Ultimate Trap

Core Rule (extremely error-prone!):

After synchronised shifting, **the remainder's decimal point = the dividend's original decimal point position**

E.g.: $1.5 \div 0.4 \rightarrow \text{synchronised } \times 10 \rightarrow 15 \div 40$

$15 \div 40 = 0 \text{ remainder } 15$ (but this is the shifted remainder!)

Restore: remainder 15 must be $\div 10 \rightarrow$ the true remainder is **1.5**

Or calculate directly by column method: $1.5 \div 0.4 = 3 \text{ remainder } 0.3$

Check: $0.4 \times 3 + 0.3 = 1.2 + 0.3 = 1.5 \checkmark$

Catastrophe Example

Ex5: $2.3 \div 0.6 = ?$ (Find the quotient and remainder)

Catastrophe (85% of students wrong)

$$2.3 \div 0.6 = 3 \text{ rem } 5$$

After synchronised $\times 10$: $23 \div 6 = 3 \text{ rem } 5$, forgot to $\div 10$ to restore the remainder!

Correct Answer

$$2.3 \div 0.6 = 3 \text{ rem } 0.5$$

Check: $0.6 \times 3 + 0.5 = 1.8 + 0.5 = 2.3 \checkmark$

Example

Ex6: $3.7 \div 0.8 = ?$ (Find the quotient and remainder)

Mnemonic: "After shifting and dividing, restore the remainder; the remainder follows the original decimal point. Verify by multiplying quotient by divisor and adding the remainder — without checking, disaster strikes."

(Cantonese: 「移位計完要還原，餘數跟返原小數點；驗算乘除加餘數，唔驗算實出事」)

Catastrophe: After synchronised shifting, the remainder must be restored! When shifting by $\times 10 / \times 100$, the remainder must be $\div 10 / \div 100$ to restore!

KP3 Synchronised Practice (Find quotient AND remainder for each; must verify)

#	Question	Difficulty	Working
11	$1.7 \div 0.5 = ?$ (Find quotient and remainder)		
12	$2.5 \div 0.7 = ?$ (Find quotient and remainder)		
13	$4.3 \div 1.2 = ?$ (Find quotient and remainder)		
14	$5.8 \div 1.6 = ?$ (Find quotient and remainder)		
15	$3.2 \div 0.9 = ?$ (Find quotient and remainder)		

KP4: Application Problems SSPA

Four-Step Problem-Solving Method: Circle keywords → Write the expression → Calculate (shift + column method) → Write a complete answer statement

Pay attention to unit consistency — dollars/cents, metres/centimetres must be unified before calculation.
Pay attention to the practical meaning of the remainder — "how much money remains", "how many metres are left"

Example

Ex7: A ribbon is 6.4 m long. It is cut into pieces of 0.8 m each. What is the maximum number of pieces? How many metres are left?

Example

Ex8: Mum has \$50. Each bag of sweets costs \$3.50. What is the maximum number of bags she can buy? How much money is left?

Application Practice

#	Question	Difficulty	Working
16	A rope is 9.6 m long. It is cut into pieces of 2.4 m each. How many pieces can be cut?		
17	Siu Ming has \$20. Each pencil costs \$3.20. What is the maximum number of pencils he can buy? How much money is left?		

III. Tiered In-Class Practice

Basic Tier (6 questions, all students must complete)

#	Question	Difficulty	Working
18	$6.4 \div 2 = ?$		
19	$8.1 \div 3 = ?$		
20	$4.5 \div 5 = ?$		
21	$2.4 \div 0.4 = ?$		
22	$6.3 \div 0.9 = ?$		
23	$3.6 \div 1.2 = ?$		

Advanced Tier (6 questions, optional)

#	Question	Difficulty	Working
24	$9.75 \div 2.5 = ?$		
25	$14.4 \div 3.6 = ?$		
26	$0.84 \div 0.12 = ?$		
27	$1.26 \div 0.3 = ?$		
28	$2.8 \div 0.7 = ?$ (Find quotient and remainder)		
29	$3.5 \div 0.6 = ?$ (Find quotient and remainder)		

Challenge Tier (5 questions, SSPA killer questions)

#	Question	Difficulty	Working
30	$5.76 \div 1.8 = ?$		
31	$0.056 \div 0.007 = ?$		
32	$4.7 \div 1.3 = ?$ (Find quotient and remainder; quotient to be an integer)		
33	A ribbon is 15.6 m long. Each bow requires 2.4 m of ribbon. What is the maximum number of bows that can be made? How many metres of ribbon are left?		

34	$3.85 \div 1.1 = ?$ (Find the quotient correct to 1 decimal place, and state the remainder)		
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IV. Application Problem Drill (SSPA Word Problems)

#	Question	Difficulty	Working
35	A bottle of orange juice contains 2.4 L. It is shared equally among 6 people. How many litres of orange juice does each person get?		
36	There are 5.5 kg of rice. The rice is packed into bags of 0.5 kg each. How many bags can be filled?		
37	The area of a rectangular piece of paper is 18.6 cm^2 . Its length is 6.2 cm. What is its width in cm?		
38	Dad drove for 3.5 hours and travelled 227.5 km. What was his average speed in km per hour?		
39	The perimeter of a rectangular garden is 18.4 m. Its width is 3.6 m. What is its length in m?		
40	Siu Mei spent \$38.50 on 5.5 kg of apples. What is the cost per kg of apples?		

V. Ultimate Challenge Zone

#	Question	Difficulty	Working
1	Calculate: $(8.4 - 2.7) \div 1.9 = ?$ (Do the subtraction inside the brackets first, then divide)		
2	Siu Ming has \$100. He buys 3 books at \$15.80 each. With the remaining money, what is the maximum number of pens at \$3.50 each that he can buy? How much money is left?		
3	Orange juice at the supermarket: Brand A — 2.5 L for \$32.50; Brand B — 1.8 L for \$25.20. Which brand is cheaper per litre? By how much? (Give your answer correct to 1 decimal place)		

VI. Homework

Compulsory Basic Questions (6 questions — must show column working)

#	Question	Difficulty	Working
H1	$7.2 \div 3 = ?$		
H2	$9.6 \div 4 = ?$		
H3	$5.6 \div 0.8 = ?$		
H4	$3.6 \div 1.2 = ?$		
H5	$4.5 \div 0.9 = ?$		
H6	$2.8 \div 0.7 = ?$ (Find quotient and remainder)		

Advanced Optional Questions (3 questions, optional)

#	Question	Difficulty	Working
H7	$8.64 \div 2.4 = ?$		
H8	A rope is 12.8 m long. It is cut into pieces of 3.2 m each. How many pieces can be obtained?		
H9	$3.85 \div 0.55 = ?$ (Hint: the divisor has 2 decimal places; multiply both by 100)		

VII. Core Error Summary for This Lesson

Learning Objectives Review — By the end of this lesson you should be able to:

- Identify all trap types in this lesson
- Independently solve Basic questions (100% correct)
- Tackle Advanced questions (80%+ correct)
- Explain this lesson's mnemonics to a classmate

#	Error Point	Correct Approach
1	Misplaced decimal point in quotient: Writing $7.5 \div 5$ as 15 instead of 1.5	Align the quotient's decimal point with the dividend's decimal point
2	Shifting only the divisor: For $3.6 \div 1.2$, only multiplying 1.2 by 10	Both dividend and divisor must be shifted synchronously (multiplied by the same factor)
3	Forgetting to restore the remainder after shifting (catastrophe): Writing $2.3 \div 0.6$ as 3 rem 5	Divide the remainder by the shift factor to restore it, or verify by checking
4	Forgetting to write "0" in the quotient when the digit is too small: For $1.2 \div 5$, the quotient should be 0.24, not 24	Every place must have a quotient digit; write 0 when the dividend digit is too small
5	Not appending 0 to continue dividing: $7 \div 4 = 1.75$, not 1 rem 3	When there is a remainder, append 0 and continue dividing until the division terminates or the specified number of decimal places is reached

6	Forgetting unit conversion in application problems: Mixing dollars and cents, metres and centimetres	Unify all units before calculation
7	Missing answer statement / missing remainder in application problems: Only writing the quotient, not "how much remains"	A complete answer statement includes both the quotient and the remainder; otherwise method marks are deducted